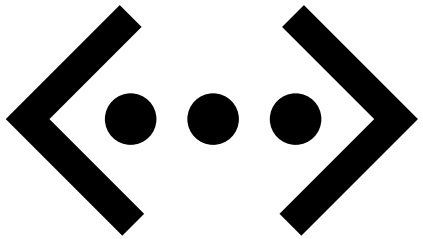
Hybrid Quantum – Classical Search

Ufuk, Nick, Samantha



- Not really too efficient but gets job done
- Classical to minimize cost (hoping for =0), and quantum to solve shots
- Bonus – 8 cubes (24 qubits) ✓
- Thank you Claude



4 cubes

What it prints (seed 7)

At N=4 the chance that one shot is a valid 2-factor, and the shots needed to assemble a verified tower:

Sampler	P(valid per shot)	Median shots	95th percentile
QAOA p=2	28.4%	17	51
QAOA p=3	44.2%	10	30
Grover, 3 rounds	about 100%	5	15

With an unknown number of solutions, BBHT needs about 6.6 oracle calls on average to get one valid 2-factor. Scaling, shots to see both needed 2-factors with 95% confidence:

N	QAOA p=3	Grover
5	127	21
6	130	13
7	fails	13

4 cubes

```
seed 7

================================================================================
N=4 starter puzzle
================================================================================
classical brute force : 331,776 stackings, 8 solved, 0.11 s
classical graph method: 81 front/back picks → 4 2-factors → tower found, 0.3 ms, verified True

Quantum stage A: constrained QAOA, 12 qubits (simulated)
  QAOA p=2: P(valid 2-factor per shot) 28.4% (random valid pick 4.9%)
      shots to a verified tower over 1000 runs: median 18, mean 22, 95th pct 51
  QAOA p=3: P(valid 2-factor per shot) 44.2% (random valid pick 4.9%)
      shots to a verified tower over 1000 runs: median 11, mean 13, 95th pct 31

Quantum stage B: Grover, 12 qubits, ideal oracle (simulated)
  Grover k=3: P(valid 2-factor per shot) 100.0% (random valid pick 4.9%)
      shots to a verified tower over 1000 runs: median 5, mean 6, 95th pct 13
      unknown number of solutions (BBHT): 6.5 oracle calls on average to get one valid 2-factor (known-count Grover uses 3 per attempt)

Classical stage on Qiskit-style counts (decoder round trip)
  1000 Grover shots → 4 distinct 2-factors seen, 2 verified tower(s): [('B', 'G', 'R', 'Y'), ('Y', 'Y', 'G', 'B'), ('R', 'B', 'Y', 'R'),
('G', 'R', 'B', 'G')]

Reminder: the circuit's job here is sampling 2-factors; brute force and the graph method are instant at this size.
```

4 cubes

N=4 Starter Puzzle Comparison (seed 7)

1. Classical baselines

Method	Key result	Time
Brute force	331,776 stackings; 8 solved	0.11 s
Graph method	81 front/back picks → 4 candidate 2-factors → tower found; verified True	0.3 ms

2. Quantum samplers

At N=4, the chance that one shot is a valid 2-factor, and the shots needed to assemble a verified tower:

Sampler	P(valid 2-factor per shot)	Median shots	Mean shots	95th percentile
QAOA p=2	28.4%	18	22	51
QAOA p=3	44.2%	11	13	31
Grover k=3	100.0%	5	6	13

Random valid pick baseline: 4.9%.

8 cubes

```
seed 7

================================================================================
N=8 generated puzzle (seed 4)
================================================================================
classical brute force : 110,075,314,176 stackings, skipped (too many to enumerate)
classical graph method: 6561 front/back picks → 6 2-factors → tower found, 41.4 ms, v
erified True

Quantum stage A: constrained QAOA, 24 qubits (simulated)
  QAOA p=2: P(valid 2-factor per shot) 3.9% (random valid pick 0.1%)
      shots to a verified tower over 1000 runs: median 174, mean 186, 95th pct inf  (58
runs found none in 500 shots)
  QAOA p=3: P(valid 2-factor per shot) 6.2% (random valid pick 0.1%)
      shots to a verified tower over 1000 runs: median 98, mean 119, 95th pct 295  (4 r
uns found none in 500 shots)

Quantum stage B: Grover, 24 qubits, ideal oracle (simulated)
  Grover k=25: P(valid 2-factor per shot) 99.9% (random valid pick 0.1%)
      shots to a verified tower over 1000 runs: median 8, mean 9, 95th pct 21
      unknown number of solutions (BBHT): 52.8 oracle calls on average to get one valid
2-factor (known-count Grover uses 25 per attempt)

Classical stage on Qiskit-style counts (decoder round trip)
  1000 Grover shots → 6 distinct 2-factors seen, 2 verified tower(s): [('E', 'G', 'G',
'F'), ('H', 'A', 'D', 'C'), ('A', 'B', 'C', 'G'), ('F', 'D', 'E', 'B'), ('B', 'H', 'A',
'A'), ('D', 'F', 'F', 'E'), ('G', 'E', 'B', 'H'), ('C', 'C', 'H', 'D')]

Reminder: the circuit's job here is sampling 2-factors; brute force and the graph metho
d are instant at this size.
```

8 cubes

```
  qubits                                          24   (3 per cube, one-hot)
  Grover iterations (known count)                 25
  BBHT oracle calls (unknown count)             52.8   (mean, to one 2-factor)
  note: ideal oracle, so oracle ancilla qubits and gates are not counted here

Success rates
  method              P(2-factor)      vs  solved in  median  95th pct  solved in  solved in
                         per shot  random  500 shots   shots     shots   10 shots   50 shots
  random valid pick          0.1%    1.0x       0.7%    none      none       0.0%       0.0%
  QAOA p=2                   3.9%   42.6x      94.2%     174      none       0.8%       9.8%
  QAOA p=3                   6.2%   68.1x      99.6%      98       295       1.8%      21.7%
  Grover k=25               99.9% 1092.6x     100.0%       8        21      70.9%     100.0%
```

Questions!